Advanced Quantitative Portfolio Management

From factor research to a portfolio you can defend in front of an investment committee.

FORMAT	DURATION	EFFORT	CPD HOURS
Online · live + self-paced	6 weeks	≈ 6 hours / week	36
LIVE SESSIONS	**NEXT COHORT**	**TOOLS**	**CERTIFICATE**
Tuesdays 7:00–8:30 pm HKT	Starts 23 November 2026	Python · pandas · cvxpy · scikit-learn	FinEx certificate of completion

OVERVIEW

Most quantitative courses stop at the model. This programme follows an idea all the way to the trade blotter: where an expected return comes from, how risk is really measured, why the 'optimal' portfolio is so often the worst one out of sample, what a backtest can and cannot prove, and how to explain the result to someone who has to sign off on it.

OUTCOMES

- ✓ Derive and critique mean-variance, minimum-variance, risk-parity and Black–Litterman portfolios
- ✓ Build, neutralise and combine factor signals into a single expected-return forecast
- ✓ Estimate a covariance matrix that survives out of sample, using shrinkage and factor models
- ✓ Run a backtest that accounts for costs, turnover and multiple testing
- ✓ Decompose performance into allocation, selection and factor contributions
- ✓ Present a portfolio and its risks to an investment committee with conviction

SYLLABUS

W1 Portfolio foundations & factor investing — 6 h
Mean–variance: the idea that started it all · CAPM, factors and the cross-section of returns · The fundamental law of active management

W2 Building a research workflow in Python — 6 h
Data you can trust: point-in-time, survivorship and look-ahead · From raw characteristic to clean signal · Reproducible research

W3 Risk models and portfolio construction — 6 h
The estimation-error problem · Shrinkage and factor risk models · Risk budgeting and risk parity · Black–Litterman: blending equilibrium with views

W4 Backtesting and execution strategies — 6 h
Backtest overfitting and the deflated Sharpe ratio · Transaction costs and market impact · Optimal execution · Trading in Hong Kong: market structure for quants

W5 Performance attribution — 6 h
Brinson attribution · Factor-based attribution and risk-adjusted metrics

W6 Final portfolio case study — 6 h
Building the capstone portfolio · The investment committee

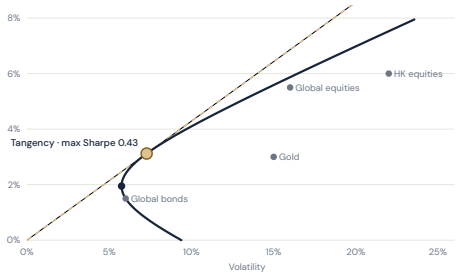
CAPSTONE

An APAC multi-factor equity mandate

You are handed a HK$2bn Asia-Pacific equity mandate benchmarked to a regional index, with a 4% tracking-error budget and a 30% annual turnover limit. Build a value-momentum-quality portfolio, size it with a risk model you can justify, backtest it honestly, and present it to a mock investment committee.

- ✓ A reproducible research notebook (signal → portfolio → backtest)
- ✓ A two-page investment memo with the risk budget and expected tracking error
- ✓ A 10-minute committee presentation with live Q&A

The efficient frontier (Week 1)



Interactive on the course page: move risk aversion and watch the optimal portfolio move.

FACULTY

Dr. Marcus Vance

Former Chief Investment Officer · CFA · Quantitative investing

An illustrative faculty profile focused on translating investment research into practical portfolio decisions, with hands-on modelling and real-world case discussions.

FEES

HK$2,800

Members HK$2,240 · 36 CPD hours · next cohort 23 November 2026

Enrol online: FinEx Careers → Learning

KEY RESEARCH

Markowitz, H. (1952). Portfolio Selection. Journal of Finance, 7(1), 77–91.

Fama, E. & French, K. (1993). Common risk factors in the returns on stocks and bonds. Journal of Financial Economics, 33(1), 3–56.

Asness, C., Moskowitz, T. & Pedersen, L. (2013). Value and Momentum Everywhere. Journal of Finance, 68(3), 929–985.

Harvey, C., Liu, Y. & Zhu, H. (2016). …and the Cross-Section of Expected Returns. Review of Financial Studies, 29(1), 5–68.

Grinold, R. & Kahn, R. (2000). Active Portfolio Management (2nd ed.). McGraw-Hill.

Jegadeesh, N. & Titman, S. (1993). Returns to Buying Winners and Selling Losers. Journal of Finance, 48(1), 65–91.

Novy-Marx, R. (2013). The other side of value: The gross profitability premium. Journal of Financial Economics, 108(1), 1–28.

Kenneth R. French Data Library, Tuck School of Business, Dartmouth College (online).

Michaud, R. (1989). The Markowitz Optimization Enigma: Is 'Optimized' Optimal? Financial Analysts Journal, 45(1), 31–42.

Ledoit, O. & Wolf, M. (2004). Honey, I Shrunk the Sample Covariance Matrix. Journal of Portfolio Management, 30(4), 110–119.

DeMiguel, V., Garlappi, L. & Uppal, R. (2009). Optimal Versus Naive Diversification. Review of Financial Studies, 22(5), 1915–1953.

Black, F. & Litterman, R. (1992). Global Portfolio Optimization. Financial Analysts Journal, 48(5), 28–43.

Illustrative programme. Faculty profile, dates and fees are sample content; CPD hours are FinEx learning hours and do not imply accreditation by any regulator or professional body.